Chatbot Development (RAG-29)

[RAG-15] Action: Set Up Project Environment & Dependencies

Created: 10/Aug/26 Updated: 10/Aug/26 Resolved: 10/Aug/26

Status:	Done
Project:	Chatbot Development
Components:	None
Affects versions:	None
Fix versions:	None
Parent:	Chatbot Development

Type:	Task		
Reporter:	Amal S	**Assignee:**	Unassigned
Resolution:	Done	**Votes:**	0
Labels:	None		
Remaining Estimate:	0 minutes		
Time Spent:	0 minutes		
Original estimate:	0 minutes		

Rank:	0|i0004n:
Sprint:	SPRINT 1

Description

Initialize the local Git repository, configure the Python virtual environment (venv/conda), and set up the base requirements.txt file. Install core required libraries including langchain/llamaindex, streamlit, chromadb, and ollama. Ensure a clean, reproducible local environment. (Compliance: K1, P5)

Objective
Initialize a robust, secure, and reproducible polyglot workspace for the Local RAG Chatbot prototype, adhering to modern software engineering standards.

Technical Architecture & Directory Structure
The workspace follows a clean separation of concerns between the Java enterprise backend, Python AI engine, local vector store, and documentation assets under a root directory:

```
RAG Chatbot/
├── data/
│   └── sops/ # Local repository for dummy SOP PDFs (IT Policy, Lab Safety)
├── docs/ # Project guidance, functional specs, and requirements
├── evidence/ # Jira screenshots and milestone compliance logs
├── java-orchestrator/ # Spring Boot backend (File scanning, orchestration, API routing)
├── python-ai-engine/ # FastAPI server, LangChain text chunking, and ChromaDB integration
├── .gitignore # Excludes venv/, .DS_Store, and temporary cache files
└── README.md # Project onboarding and architecture documentation
```

Future Deployment & Migration Strategy

- **Current Phase:** Local development and sandbox prototyping executed on an Apple M1 MacBook Pro utilizing local unified memory and Apple Metal GPU acceleration for Ollama.
- **Production / Target Phase:** Code will be containerized using **Podman** (rootless and daemonless containers for enhanced security compliance) and migrated to a dedicated home server (Dell OptiPlex with 32GB RAM) accessible securely via **Tailscale** zero-config VPN tunneling.

Implementation Checklist

- [x] Establish root project directory and sub-modules on local machine
- [x] Initialize Git repository with secure `.gitignore` rules
- [x] Configure Python 3.9+ virtual environment (`venv`) and install core dependencies (`langchain`, `chromadb`, `fastapi`, `streamlit`, `ollama`)
- [x] Compile and lock package dependencies into `requirements.txt`
- [x] Set up foundational Spring Boot project structure in IntelliJ IDEA (JDK 21, Maven)

Comments

Comment by Amal S [10/Aug/26]

Status Update: Completed. Initialized master workspace structure, configured Python virtual environment with all required AI and web packages, established Git tracking, and mapped out the transition roadmap from local M1 MacBook development to containerized Podman deployment on the home server.